Ladder Explained

Now this is the right place to slow down and really understand what the ladder test actually is, why it exists, and how to read it without fooling yourself. I'll go deep, but keep it practical and grounded in real-world shooting, not internet mythology.

What the Ladder Test Really Is (and Why It Exists)

At its core, a ladder test is not about group size**.** **A ladder test is a pressure-vs-velocity-vs-barrel timing diagnostic.**

It exists to show you where small increases in powder stop moving point of impact vertically. That's it.

Everything else people claim it does is secondary. The Physics Behind the Ladder (why it works) When you increase powder charge:

- Velocity increases
- Barrel time changes
- Barrel whip timing changes
- Bullet exits the muzzle at a different phase of barrel vibration

This causes:

Vertical POI shift, even with perfect shooting

Key concept: barrel harmonics

Every barrel vibrates like a tuning fork when fired. There are moments in the vibration cycle where:

- The muzzle is moving very little vertically
- Bullet exit timing is more forgiving

These moments create "nodes".

A ladder test helps you find charge weights where velocity increases, but vertical **POI** does NOT. **That is the magic.**

What a Ladder Test Is NOT

Let's kill some myths:

❌ It is NOT for finding the smallest group

❌ It is NOT a replacement for OCW

❌ It is NOT about SD

❌ It is NOT reliable at 100 yd (unless you know what you're doing)

A ladder is about vertical trends, not clusters.

Why Distance Matters (a lot) - Vertical dispersion grows with distance.

At 100 m / yd → velocity differences hide

At 300+ m / yd → nodes become obvious

At 600 m / yd → they scream at you

Ideal ladder distances:

300 yd minimum

400–600 yd preferred

This is why ladder tests were originally a long-range benchrest tool, not a 100-m / yd range trick.

How a Proper Ladder Test Is Shot

Load prep:

- Same brass lot
- Same primer
- Same seating depth
- Charges usually in 0.2–0.3 gr increments
- One round per charge

Example:

41.0, 41.3, 41.6,41.9,42.2,42.5

Shooting method:

- One aiming point
- Shoot low to high charge
- Let barrel cool between shots
- Same wind condition if possible

Each shot:

- Different charge
- Different velocity
- Different barrel timing

How to Read Ladder Results (this is the heart of it). After shooting, ignore horizontal dispersion.

You care about vertical only.

You'll typically see:

- Shots climbing steadily
- Then two or more shots land at nearly the same vertical height
- Then climb resumes

Example (vertical only):

42.5 | o

42.2 | o

41.9 | o o <-- NODE

41.6 | o o <-- NODE

41.3 | o

41.0 | o

That flat spot = **Likely a node**

What it means:

- Velocity increased
- Barrel timing stayed favorable
- Bullet exited at same muzzle position
- That charge region is inherently forgiving.

Why Flat Vertical = Forgiveness

Let's say:

You load 41.9 today.

- Next month it's hotter
- Pressure rises
- Velocity increases slightly

If you're in a node:

- POI barely changes

If you're between nodes:

- POI shifts dramatically

That's why ladder-found nodes: **Hold vertical at distance, Track temperature better**

Don't "fall apart" in matches

Why One Round Per Charge Is Enough

This confuses people.

You're not averaging anything. You're not grouping. You're mapping a trend. Think of it like plotting a graph:

X-axis = charge and Y-axis = vertical POI

You don't need repeats to see slope changes. Pressure Signs During the Ladder. This is also where ladders shine

As you climb charges:

- Bolt lift
- Primer flattening
- Ejector marks
- Velocity jumps
- If pressure spikes:

Stop immediately

Node above it is irrelevant

You've just defined your safe ceiling.

Ladder vs OCW (why ladder comes first)

Ladder test answers:

"Where are the barrel timing nodes and safe pressure limits?"

OCW answers:

"Which of these nodes is the most repeatable in group form?"

You use ladder results to:

- Narrow powder range
- Eliminate bad regions
- Avoid wasting OCW components

Common Ladder Mistakes (and why people say they "don't work")

- Shooting at 100 yd
- Shooting multiple shots per charge
- Chasing group size
- Shooting too fast (heat shifts POI)
- Ignoring wind vertical
- Using inconsistent brass

When a Ladder Test Is Most Useful

✓ New powder

✓ New bullet weight

✓ Unknown pressure window

✓ Long-range rifle

✓ Limited components

If you already know the system well, OCW alone can be enough — but ladder gives you context and safety.

How Ladder Results Feed the OCW

After the ladder:

You'll usually see 1–2 viable nodes in green in the App. Pick one (often lower for safety)

Run OCW inside that node

Example:

Ladder node at 41.6–42.0

OCW charges: 41.6 / 41.8 / 42.0

Now OCW has a much higher success rate.

Final Mental Model (this is the key takeaway)

Ladder = maps barrel behaviour

OCW = validates stability

Seating depth = sharpens precision

SD = consistency, not accuracy

One-sentence ladder summary:

A ladder test reveals charge weights where increasing velocity stops moving vertical POI, identifying forgiving barrel timing nodes before you ever worry about group size.